

Kusuma Kumari Penta

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EDUCATION

Indian Institute of Technology, Bombay, India

Jul 2019 – May 2023

Bachelor of Technology in Computer Science and Engineering with *Honors*

CPI: 8.34/10

- **Key Courses:** Operating systems, Computer Networks, Computer Architecture, Implementation of Programming languages, Database and Information systems, Topics in Virtualisation and cloud computing, Foundations of Network Security and cryptocurrency, Principles in Data and Systems security

PROFESSIONAL EXPERIENCE

Oracle Cloud Infrastructure (OCI)

Bengaluru, India

Software Engineer | File Storage Service (OCI FSS)

Jun 2023 - present

- Transitioned FSS from static NVMe disks to **dynamic, network-attached block storage volumes**, reducing storage footprint and cost by **60%** by eliminating redundant data replication and improving storage scalability.
- Implemented a custom **lease coordination protocol** using **Paxos** and **iSCSI Persistent Reservations** to ensure high availability and prevent data corruption during volume movement across storage nodes.
- Built multi-threaded worker pools in **C** with registered callbacks to orchestrate lease logic and propagate state across nodes using fine-grained mutexes, state machines, and global generation counters.
- Designed and integrated **RPC interfaces** for dynamic volume operations and contributed to core C libraries, including reference counters, memory pool allocators, volume record tables, and SCSI-PR (via ioctl calls).
- Developed **Go routines** to automate volume attachment and rebalancing across nodes by coordinating with Minsk (key-value store), OCI Block Storage APIs, and storage RPCs, with robust retry and failure-handling mechanisms.
- Improved dynamic-volume performance through **kernel buffer** usage and **in-transit encryption** experiments between storage nodes and network-attached devices, enabling fine-tuned **congestion control** across the storage stack.
- Enhanced FSS's testing infrastructure by creating low-cost VM clusters using OCI APIs in Go, reducing resource consumption and test time by **70%**, and optimised build pipelines through improved parallelism, reducing build duration by **40%**.
- Extending the **file system consistency checker** into a distributed background service with checkpointing and metadata-key distribution, automating consistency verification and improving execution time by **65%**.

Oracle Cloud Infrastructure (OCI)

Bengaluru, India

Internship | File Storage Service (OCI FSS)

May 2022 - Jun 2022

- Enhanced system observability by adding **dynamic runtime dimensions** to metrics—identified during sampling and tracked using **concurrent hash maps** and **linked lists** in C and Java—reducing log inspection time by **80%**.
- Developed a **Python utility** to query **T2-TSDB** and filter results based on dimensions using **MQL**, streamlining debugging and Customized **Grafana dashboards** to visualize dynamic metric dimensions, enabling real-time monitoring.

RESEARCH & KEY PROJECTS

Java Execution Optimization

IIT Bombay, India

R&D - prof. Varsha Apte [report]

Jan 2023 – Apr 2023

- Designed an optimized **Java evaluation subsystem** for BodhiTree's auto-grader by reusing active **JVMs** through **JShell**, reducing program evaluation latency by over **50%** and eliminating per-testcase JVM startup overhead.
- Implemented asynchronous, **Unix pipe-based IPC** to stream user programs and inputs into persistent JShell instances, enabling parallel evaluation and improving throughput by reducing request-handling latency.
- Leveraged the Linux **/proc** filesystem to monitor per-process CPU and memory usage in real time, integrating polling and runtime-restriction mechanisms for sandbox safety under varying workloads.
- Deployed and benchmarked the optimised execution pipeline on a **QEMU-assisted ARM64 Linux VM**, analysing scalability across varying numbers of active JShells and identifying optimal concurrency levels for the auto-grader.

Lock-free Queues

Bengaluru, India

Self Project

Jun 2024 - Jul 2024

- Designed and implemented **lock-free SPSC** queue library in **C++23** for fixed-width types, optimised with branch-free push/pop operations and concurrent data structures, outperforming the widely used `rigtorp::SPSCQueue` in throughput.
- Implemented multiple queue variants, including basic, no-modulo, release-acquire pairs, no false-sharing, and local-cache optimised versions, achieving up to **390M** operations per second on **x86** systems.
- Conducted cross-platform benchmarking on **ARM64** (macOS) and **x86-64** (Linux) with careful thread pinning, cache-line alignment, and detailed throughput analysis to study architectural performance divergence.

Computer Architecture

Course Project

IIT Bombay, India

Oct 2021 - Dec 2021

- Implemented an **IP-based LLC bypassing** mechanism inspired by the MadCache paper to reduce cache pollution from low-reuse data streams, achieving up to **40% IPC improvement** under memory-intensive workloads.
- Developed a **prefetch throttling mechanism** for an IP-stride prefetcher based on prefetch accuracy and coverage, optimising bandwidth utilisation and overall system performance.
- Integrated and benchmarked both mechanisms on the **ChampSim microarchitectural simulator**, analysing cache hit rate, prefetch efficiency, and IPC across diverse SPEC CPU workloads.

Operating Systems

Course Project

IIT Bombay, India

Aug 2021 - Nov 2022

- Implemented a Linux-like **command shell** in C supporting serial, parallel, and background execution using **fork**, **exec**, and **wait** system calls, with built-in commands, robust signal handling, and improved error reporting.
- Extended the **xv6** kernel with custom system calls and process management utilities to explore scheduling, context switching, and process state transitions; implemented page-table inspection and a simplified **mmap()** for virtual memory management.
- Developed a simplified **file system (SimpleFS)** in xv6 supporting file creation, read/write, and directory traversal through inode-based metadata management, block allocation mechanisms, and basic consistency checks.

Network Protocols

Course Project

IIT Bombay, India

Feb 2021 - Apr 2021

- Implemented a custom **ARQ protocol** over **UDP** in C using socket programming to ensure reliable and ordered packet delivery.
- Simulated and analysed the impact of packet loss and network delay on throughput for **TCP variants** - Tahoe, Reno, & Vegas.
- Utilised **Wireshark** tool to capture protocol behaviour, verifying correctness and performance for varying network conditions.

Compiler Design

Course Project

IIT Bombay, India

Feb 2022 - Apr 2022

- Developed an end-to-end compiler for a custom **C-like language** supporting arrays, recursion, scope and control sequences.
- Implemented lexical analysis and parsing using **Lex & Yacc**, generating **AST**, **TAC**, & **RTL** & translating into MIPS assembly code.

SKILLS

Programming Languages	C, C++, Go, Python, Bash, Java, Rust, JavaScript, TypeScript, SQL, Assembly
Systems debug & Perf Tools	GDB, Valgrind, perf, strace, ltrace, fio, iostat, vmstat, tcpdump, Wireshark, Linux
Development & Build Tools	Git, Docker, Make, CMake, Grafana, Prometheus, LaTeX

TEACHING & MENTORSHIP

- Served as a **TA** for introductory **CS101** course, assisting students with programming concepts, lab sessions & evaluation. (2022)
- Mentored 5 juniors from the Department of Computer Science as part of the **DAMP Mentorship Program**. (2021)
- Taught basic computer literacy and digital skills to underprivileged children as part of the **NSS outreach program**. (2024)

AWARDS AND HONORS

- Received **Employee Recognition Award** within the first year of tenure at Oracle. (2024)
- Achieved **All India Rank 62** in JEE Mains among 935,000 candidates. (2019)
- Secured **All India Rank 581** in IIT JEE Advanced among 245,000 candidates. (2019)
- Received KVPY fellowship with **AIR 115** in written and interview examination. (2018)
- Ranked among Top 1% across the nation in **NSEC** and **NSEP** conducted by IAPT. (2019)

EXTRACURRICULAR

- Represented FSS team in the Oracle-wide **Storage Hackathon**, competing across all distributed storage service teams. (2023)
- Participated and won in mixed **volleyball** tournament conducted by **IITB Volleyball club** out of 10+ teams. (2022)
- Completed the **Kedarkantha** (5-day) and **Kalsubai** treks organized by IIT Bombay's Aavhan team. (2022)
- Built a **Bluetooth-controlled bot** as part of the **Robotics Club**, designed for precise runway navigation. (2019)
- Served as an organiser for IIT Bombay's **Mood Indigo**, Asia's largest college cultural festival. (2019)
- Completed a course under IIT Bombay's National Service Scheme focused on sustainable social development, contributing to the development of a **plastic shredder machine** on campus for improved plastic waste processing. (2019)